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FACULTY OF INFORMATION TECHNOLOGY

Operating System

Lab No 2

C ++Programming
And
GNU Debugger

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Basic C programming in Linux Enviornment

Topics to be covered

a. **Compiling and Running a C Program.**

b. **Data Type**

c. **Input/output**

d. **Control structures (If/else/switch)**

e. **Looping**

f. **Functions with/without parameters**

g. **Command Line Arguments**

h. **Dynamic memory allocation**

i. **Structures**

Objectives:

- Students able to understand the concept of C++ Programming in Linux Enviornment.

Pre-requisite:

- GCC installed in the system
- Visual studio code must be installed.

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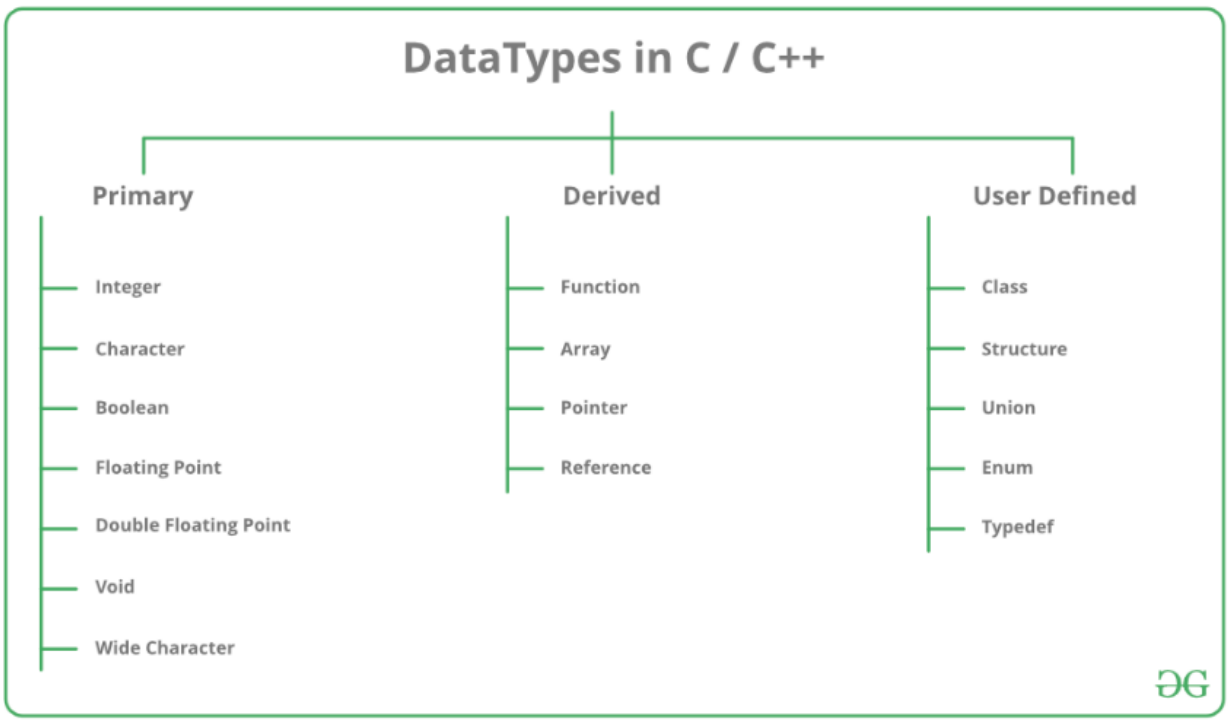
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Basic C Programming

Compiling and running a c program in Linux.

1. **Open a file using any editor e.g. gedit hello.c**
2. **Write Helloworld programs**
3. **Compile using gcc**
    a. **g++ hello.cpp and run output file (./a.out) or**
    b. **g++ hello.cpp –o myexe and run ./myexe**

Data Type



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Input/output

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CP Task 1 (0.5 Marks)

Write a C++ program that ask the user to enter a length and width of a rectangle. It calculates the rectangle area and display it on screen.

Area of rectangle = length * width

Sample Output:

What is the length of rectangle? 10
What is the width of rectangle? 20
The area of rectangle is 200.

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Control Structures

C++ if Statement

The syntax of the `if` statement in C programming is:

```
if (test expression)
 {
 // statements to be executed if the test expression is true
 }
```

How if statement works?

The `if` statement evaluates the test expression inside the parenthesis `()`.
If the test expression is evaluated to true, statements inside the body of `if` are executed.
If the test expression is evaluated to false, statements inside the body of `if` are not executed.

C++ if...else Statement

The `if` statement may have an optional `else` block. The syntax of the `if..else` statement is:

```
if (test expression) {
 // statements to be executed if the test expression is true
}
else {
 // statements to be executed if the test expression is fals
}
```

How if...else statement works?

If the test expression is evaluated to true, statements inside the body of `if` are executed.

Statements inside the body of `else` are skipped from execution. If the test expression is

evaluated to false, statements inside the body of `else` are executed statements inside the body of `if` are skipped from execution.

CP Task 2 (0.5 Marks)

Prompt a user to enter the length of three sides of a triangle. Determine if these three sides form a valid triangle. If so than determine if the triangle is scalene, isosceles or equilateral.

Hint:

In triangle no one side can be greater than the sum of other two sides.

- Scalene: No sides of the triangle are equal to each other.
- Isosceles: Two sides of the triangle are equal.
- Equilateral: All three sides are equal.

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Looping

The syntax of the `for` loop is:

```
for (initializationStatement; testExpression; updateStatement)
{
  // statements inside the body of loop
}
```

How for loop works?

The initialization statement is executed only once. Then, the test expression is evaluated. If the test expression is evaluated to false, the `for` loop is terminated. However, if the test expression is evaluated to true, statements inside the body of `for` loop are executed, and the update expression is updated. Again the test expression is evaluated. This process goes on until the test expression is false. When the test expression is false, the loop terminates.

CP Task 03 (0.5 Marks)

Write a program to perform sorting of numbers (Bubble Sort). The program shall take an array of 20 integers as input, if user wants to enter less than 20 numbers, user shall terminate it with -99. The program shall then sort the sequence in ascending order and print both the original and sorted sequence.

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Sample Input:

Input Sequence: 15 3 2 16 7 9 12 25 5 56 8 2 -99

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Sample Output:

The entered sequence is: 15 3 2 16 7 9 12 25 5 56 8 2

Updated sequence is: 2 2 3 5 7 8 9 12 15 16 25 56

Functions

A function is a block of code that performs a specific task.

CP Task 4 (0.5)

Write a **void** `checkPrimeNumber()` function which takes input from the user, checks whether it is a prime number or not and displays it on the screen. **(2+2+1)**

Sample Output: -

Enter the number: = 66

Output: Not Prime

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Command Line Arguments

Command line argument is a parameter supplied to the program when it is invoked. Command line argument is an important concept in C programming. It is mostly used when you need to control your program from outside. Command line arguments are passed to the `main()` method.

```
int main (int argc , char ** argv)
#include <siostream.h>
int main(int argc, char **argv){
cout<<(“No of arguments”,argc);
 return 0;
}
```

CP Task 5 (0.5)

Write a C++ program to receive two integers from command line and Multiply them.

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Dynamic memory allocation

Dynamic Memory Allocation with new and delete

- **new**: Allocates memory for a single object or an array of objects.
- **delete**: Deallocates memory allocated with new.

```
#include <iostream>

int main() {
    // Allocate memory for a single integer
    int* ptr = new int;  // Dynamic allocation
    *ptr = 42;

    std::cout << "Value: " << *ptr << std::endl;

    // Deallocate memory
    delete ptr;

    // Allocate memory for an array of integers
    int* arr = new int[5];  // Dynamic array allocation
    for (int i = 0; i < 5; ++i) {
        arr[i] = i + 1;  // Assign values
    }

    std::cout << "Array: ";
    for (int i = 0; i < 5; ++i) {
        std::cout << arr[i] << " ";
    }
    std::cout << std::endl;

    // Deallocate memory for the array
    delete[] arr;

    return 0;
}
```

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CP Task 6 (0.5)

Write a C++ program to display second min number in an array. User enter integers and you need to calculate the size of array and allocate run time memory for array.

Sample Output: -

Enter the array elements: = ./a.out 1 2 3 4 5 6 7 8 9 0 Output:
1

Structures

Syntax of struct

```
#include <iostream>
#include <string>

struct Person {
    std::string name;  // Member variable for name
    int age;           // Member variable for age
    float height;      // Member variable for height
};
```

CP Task 7 (0.5)

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Write a C++ program stores the information (id, name, age and pay) of an employee and displays it on the screen using structures

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CP Task 8 (10.5)

You are given a file "task1.txt" containing record of 10 employees (Fig. 1). Write a C program to manage employee's information using structures. The information of an employee contains ID (i.e. Emp01), Name (i.e. without space), gender (i.e. m/f), job position (i.e. internee/developer without spaces), experience in years (i.e. 1/2) and pay. All the data will be saved to a file in specific format as depicted in Fig. 1. Maximum number of employees are 100. The program will prompt the user a Menu for different operations as shown below:

========== **MENU** ==========

1. Add a record
2. Search a record by ID
3. Show all records
4. Show employees having pay less then basic pay (20000)
5. Save and exit

Task1.txt

Emp01 Asad M internee 1 14000
Emp02 Ahmad M developer 2 50000
Emp03 Ayesha F sqaanalyst 5 35000
Emp04 Amjad M dataanalyst 3 30000
Emp05 Bushra F personalassistant 4 25000
Emp06 Bisma F developer 2 40000

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Fig. 1. File format (ID Name Gender Position Pay)

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Hint. You can read all the records at a time using structures array. You can update file by opening it in append mode.

Functions prototype

void printStd(employee tmp) void readData(employee rec[], int &count) void addRecord(employee rec[], int *count) void searchByID(char reg[], employee rec[], int count) void showAllRecord(employee rec[], int count)
void showBelowBasicpay(employee [], int count, int basicpay)

CP Task-08

GNU Debugger

To learn C program debugging, let us create the following C program that calculates and prints the factorial of a number. However, this C program contains some errors in it for our debugging purpose.

```
int main()
{
    int i, num, j;
    printf ("Enter the number: ");
    scanf ("%d", &num );

    for (i=1; i<num; i++)
        j=j*i;

    printf("The factorial of %d is %d\n",num,j);
}
```

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Step 1. Compile the C program with debugging option -g

Compile your C program with -g option. This allows the compiler to collect the debugging information.

```
$ cc -g factorial.c
```

Note: The above command creates a.out file which will be used for debugging as shown below.

Step 2. Launch gdb

Launch the C debugger (gdb) as shown below.

```
$ gdb a.out
```

Step 3. Set up a break point inside C program

```
Syntax:

break line_number
```

Other formats:

- break [file_name]:line_number
- break [file_name]:func_name

Places break point in the C program, where you suspect errors. While executing the program, the debugger will stop at the break point, and gives you the prompt to debug.

So before starting up the program, let us place the following break point in our program.

```
break 10

Breakpoint 1 at 0x804846f: file factorial.c, line 10.
```

Step 4. Execute the C program in gdb debugger

```
run [args]
```

You can start running the program using the run command in the gdb debugger. You can also give command line arguments to the program via run args. The example program we used here does not requires any command line arguments so let us give run, and start the program execution.

```
run

Starting program: /home/sathiyamoorthy/Debugging/c/a.out
```

Once you executed the C program, it would execute until the first break point, and give you the prompt for debugging.

```
Breakpoint 1, main () at factorial.c:10

10                      j=j*i;
```

You can use various gdb commands to debug the C program as explained in the sections below.

Step 5. Printing the variable values inside gdb debugger

```
Syntax: print {variable}



Examples:

print i

print j

print num

(gdb) p i
```

```
$1 = 1

(gdb) p j

$2 = 3042592

(gdb) p num

$3 = 3

(gdb)
```

As you see above, in the factorial.c, we have not initialized the variable j. So, it gets garbage value resulting in a big numbers as factorial values.

Fix this issue by initializing variable j with 1, compile the C program and execute it again.

Even after this fix there seems to be some problem in the factorial.c program, as it still gives wrong factorial value.

So, place the break point in 10th line, and continue as explained in the next section.

Step 6. Continue, stepping over and in – gdb commands

There are three kind of gdb operations you can choose when the program stops at a break point. They are continuing until the next break point, stepping in, or stepping over the next program lines.

- c or continue: Debugger will continue executing until the next break point.
- n or next: Debugger will execute the next line as single instruction.
- s or step: Same as next, but does not treats function as a single instruction, instead goes into the function and executes it line by line.

By continuing or stepping through you could have found that the issue is because we have not used the <= in the 'for loop' condition checking. So changing that from < to <= will solve the issue.

gdb command shortcuts

Use following shortcuts for most of the frequent gdb operations.

- l – list
- p – print
- c – continue
- s – step

- ENTER: pressing enter key would execute the previously executed command again.

Miscellaneous gdb commands

- **l command:** Use gdb command l or list to print the source code in the debug mode. Use l line-number to view a specific line number (or) l function to view a specific function.
- **bt: backtrack** – Print backtrace of all stack frames, or innermost COUNT frames.
- **help** – View help for a particular gdb topic — help TOPICNAME.
- **quit** – Exit from the gdb debugger.